

Multiple Choice (10 marks)

1. If $z = \arctan(e^{\{1 + (1+x)^2\}})$, what's the derivative of $\frac{\partial z}{\partial x}$ at $x = 0$.
 - (a) 0.4752
 - (b) 0.6210
 - (c) 0.4123
 - (d) 0.5678
 - (e) 0.5439
2. Which statement is true?
 - (a) The product of 7 x 5 is odd because both of the factors are even.
 - (b) The product of 3 x 3 is odd because one of the factors is odd.
 - (c) The product of 5 x 2 is even because both of the factors are even.
 - (d) The product of 6 x 2 is odd because one of the factors is odd.
 - (e) The product of 2 x 2 is odd because both of the factors are even.
3. 5.5-1. Let $X_1, X_2, \dots, X_{16}$ be a random sample from a normal distribution $N(77, 25)$. Compute (a) $P(77 < \bar{X} < 79.5)$.
 - (a) 0.3524
 - (b) 0.2123
 - (c) 0.5948
 - (d) 0.3187
 - (e) 0.4298
4. Find the median in the set of data 23, 13, 18, 29, 32, 25.
 - (a) 25
 - (b) 22
 - (c) 30
 - (d) 27
 - (e) 29
5. compute the integral $\int_{-\{\Gamma\}} \frac{x*dy - y*dx}{x^2 + y^2}$, where Γ is any piecewise smooth, closed curve that encloses the origin but does not pass through it.
 - (a) 3.1415926

- (b) 4.6692016
- (c) 1.5707963
- (d) 0
- (e) 2.7182818

True / False (10 marks)

Answer True or False

6. True or False: The perimeter of polygon ABCDE is 30 units, as it maintains similarity with polygon PQRS.
7. True or False: A group of 6 people can be divided into 2 ordered teams in 720 distinct ways.
8. True or False: The second derivative of $h(x)$ at $x = 1$ is equal to -0.25.
9. True or False: The expanded form of 270,240 can be expressed as $200,000 + 70,000 + 200 + 40$.
10. True or False: The norm $\|x\|_p$ can be expressed as an inner product if and only if p equals 2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Circle O is located on the coordinate plane with center at $(2, 3)$. One endpoint of a diameter is at $(-1, -1)$. What are the coordinates of the other endpoint of this diameter? Express your answer as an ordered pair.
12. Given the trajectory of a cannonball described by the function $h(t) = -4.9t^2 + 14t - 0.4$, analyze the conditions under which the cannonball remains above a height of 6 meters. Calculate the duration of this interval and discuss the implications of your findings within the context of projectile motion.

End of Paper